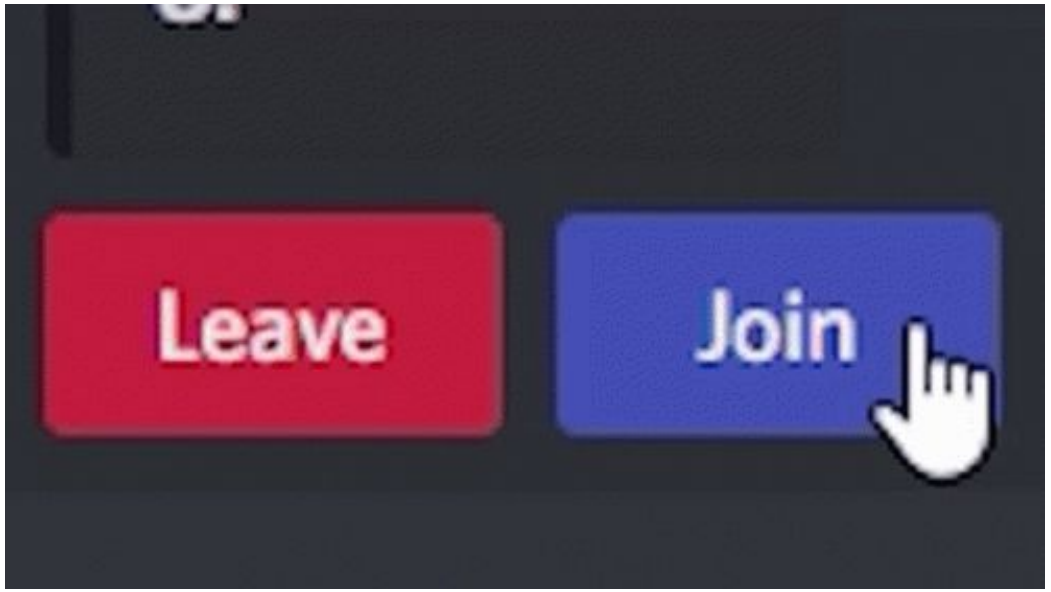
Newton School of Technology

0

Introduction to Intro to AI-ML

Join the Lecture on NS Portal :)

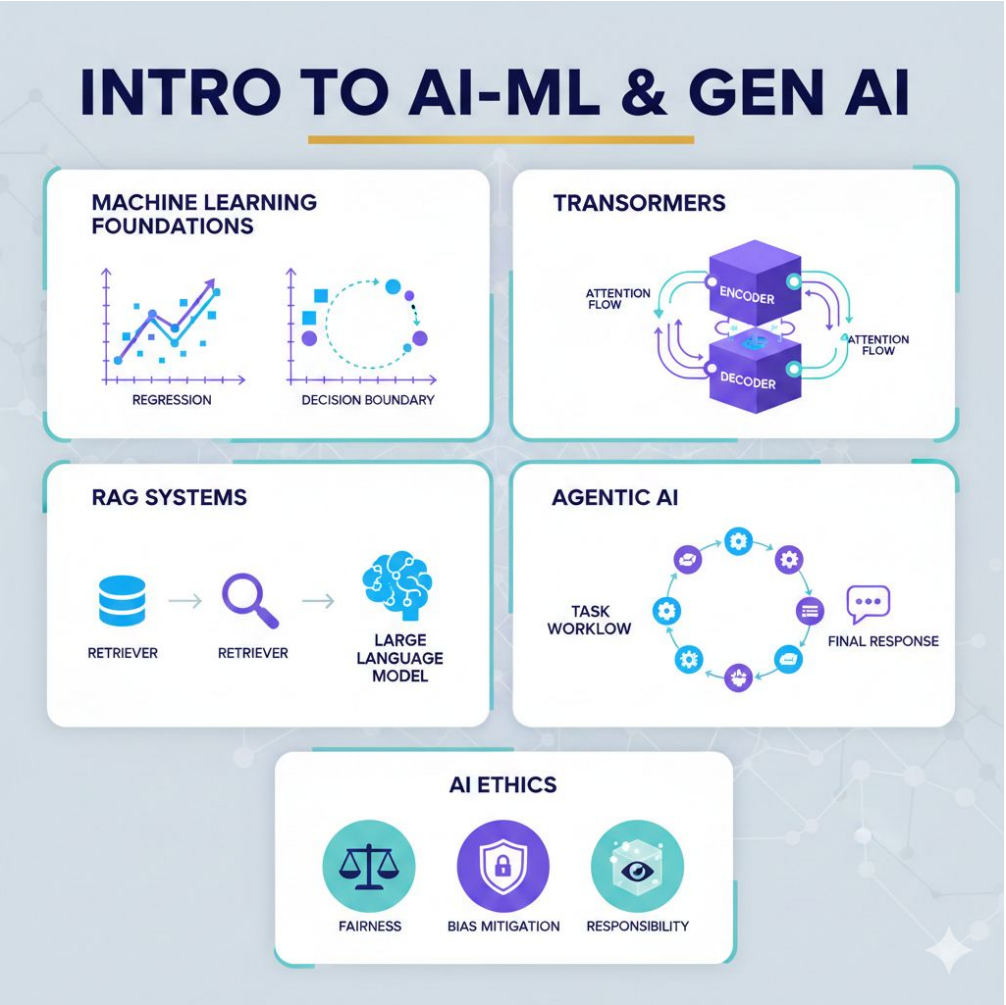
Newton School of Technology



Course Objectives

Newton School of Technology

- **Master ML Foundations: Regression & Classification basics.**
- **Transformers: Architecture and Attention mechanisms.**
- **RAG Systems: Retrieval Augmented Generation implementation.**
- **Agentic AI: Deploy autonomous workflow agents.**
- **AI Ethics: Bias mitigation and fairness.**



Your Faculties –

Instructors -

- Bipul Kumar Shahi
- Kartik Gupta

Lab Instructors -

- Ravi Kumar
- Sumit Kumar
- Arun Chauhan

What is Generative AI?

Newton School of Technology

Discriminative AI

Traditional machine learning focused on classifying or predicting based on existing data.
Example: *"Is this image a cat?"*

Generative AI

AI that creates entirely new content (text, images, code) based on learned patterns.
Example: *"Draw a cat in space."*

The 'Why' and 'How'

Newton School of Technology

Industry Shift

AI integration is no longer a luxury; it is a baseline requirement for modern engineers.

Pedagogy

40% Theory and 60% Code. We prioritize building over passive "Black Box" learning.

Collaboration

Industry-standard pair programming and rigorous peer reviews for code quality.

Expectation Setting

Newton School of Technology

✓ What to Expect

- Intuitive Understanding
- Learning to build cool AI Agents
- Awesome Projects for Resume
- Interview Readiness for AI ML Internships
- Prototyping with Streamlit
- Use of Commercial APIs

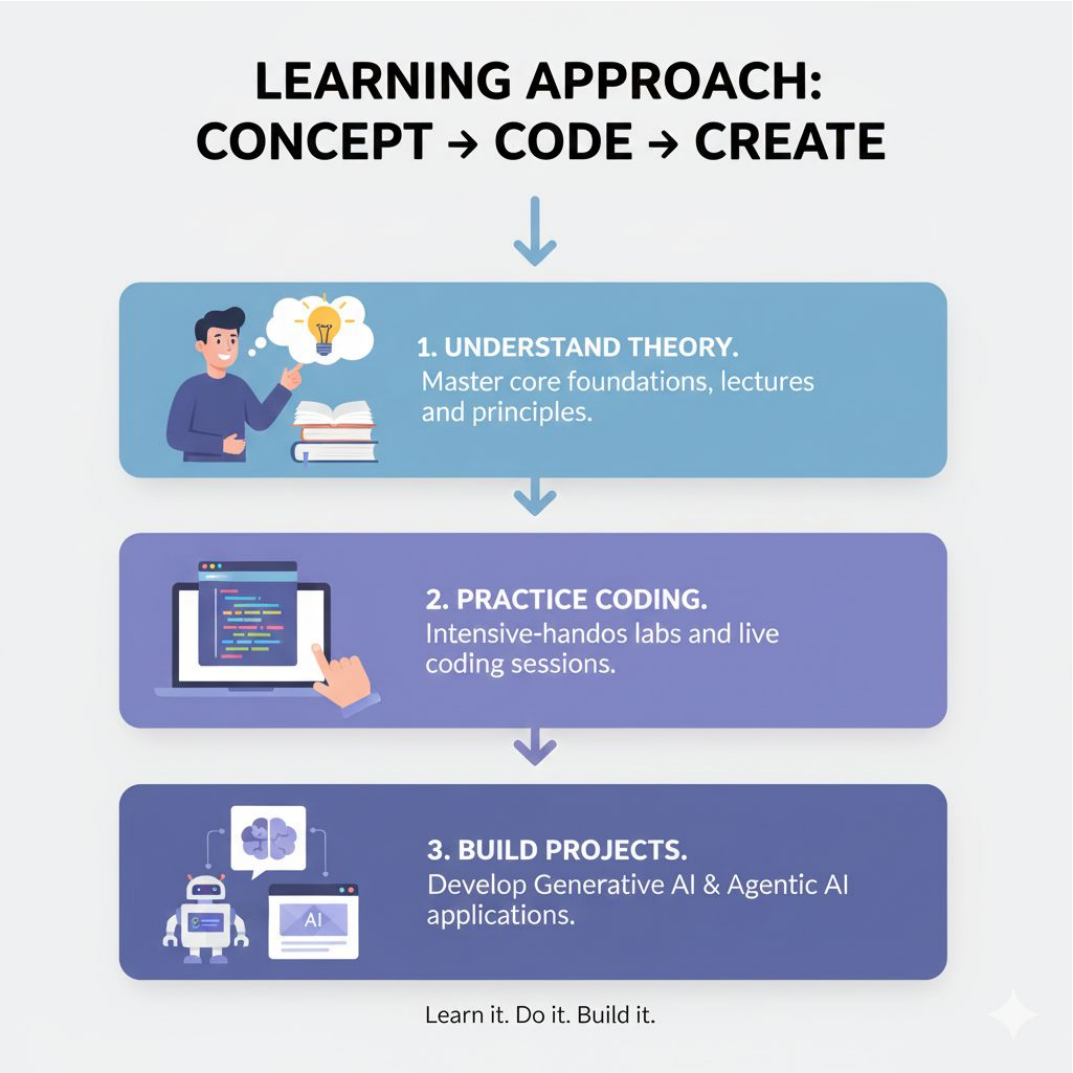
✕ What NOT to Expect

- Deep or Tough maths
- All theory with no practical
- Time to relax

Learning Process

Newton School of Technology

1. **Concept:** Lectures on core theory and foundations.
2. **Code:** Intensive live coding sessions in labs
3. **Create:** Awesome projects for gen AI and agentic AI



Career Impact

Newton School of Technology

3X
Demand Growth

82%
Industry Adoption

New Market Roles

The landscape is shifting rapidly towards specialized roles:

- AI Engineer / RAG Specialist
- Prompt Engineer
- AI Operations (AIOps)

Evaluation & Grading

Newton School of Technology

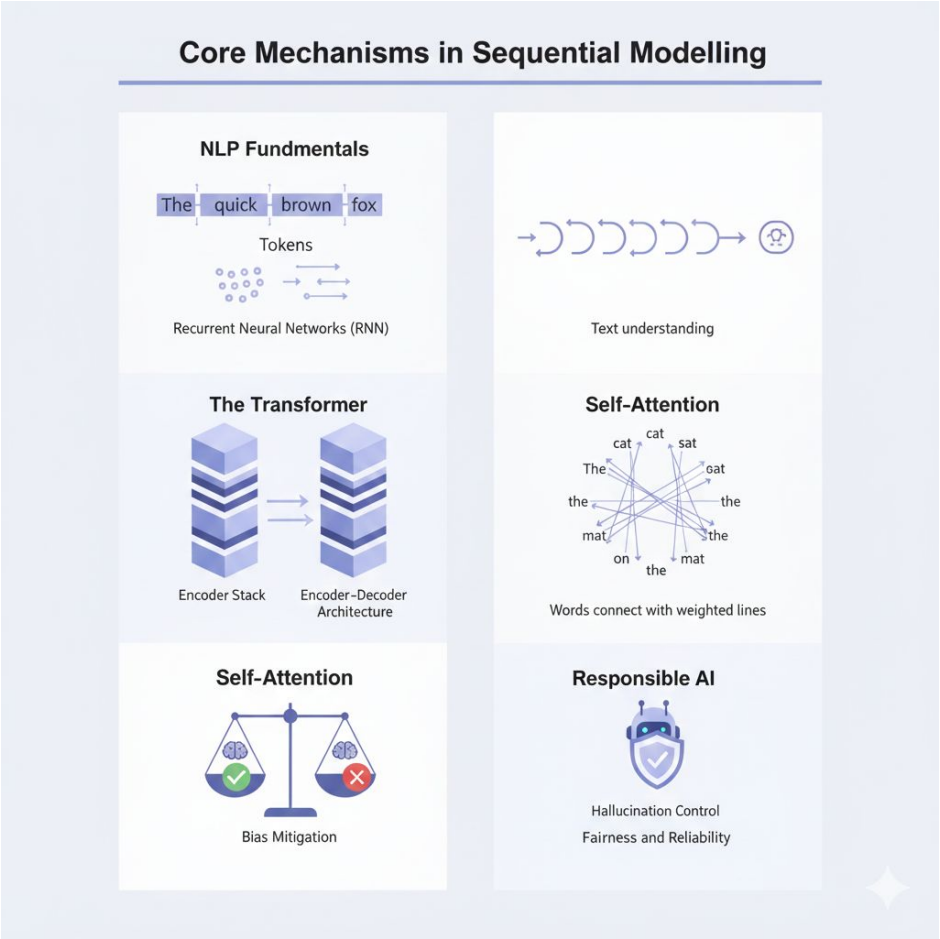
Assessment Component	Weightage	Focus Area
End Semester Exam	40%	Comprehensive Theory & Design
Mid Semester Exam	20%	Foundational Concepts
Capstone Project	20%	End-to-end GenAI Application
Contests (Pre-Mid & Pre-End)	20%	Coding Speed & Problem Solving

Phase 1: Foundations

Newton School of Technology

Module	Key Topics
Intro to ML	AI vs ML vs Deep Learning, Bias-Variance Tradeoff
Preprocessing	Handling Missing Values, Feature Scaling, Encoding
Supervised Learning	Regression, Classification, Evaluation Metrics
Neural Networks	Perceptrons, Multi-layer Backpropagation, Optimizers

Phase 2: NLP & Architecture

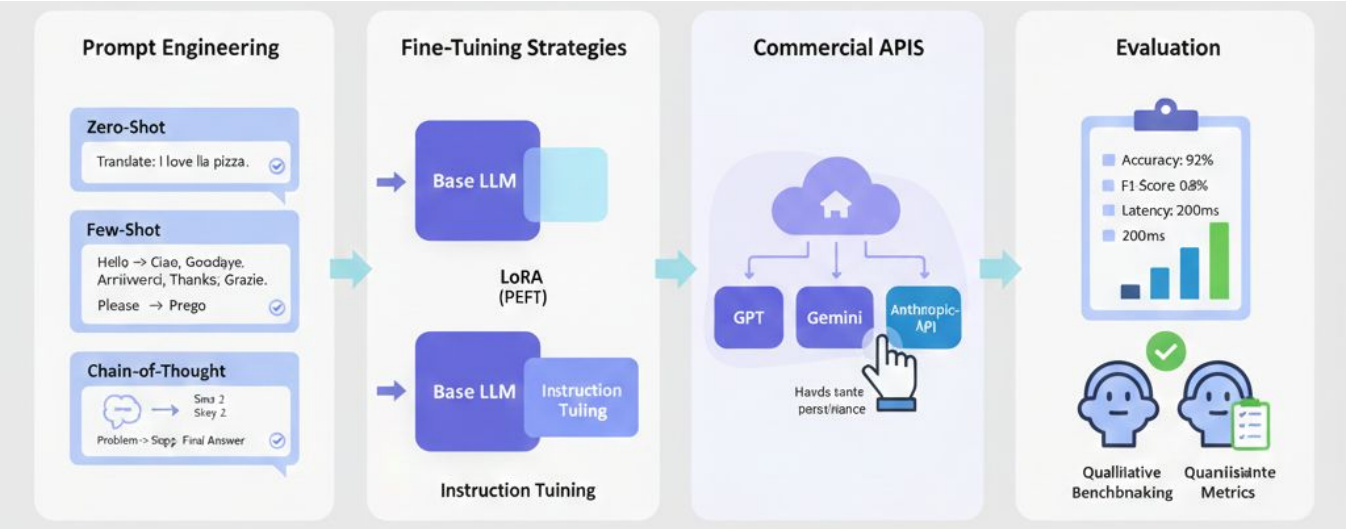


Core Mechanisms

- **NLP Fundamentals:** Tokenization, Embeddings, RNNs.
- **The Transformer:** Encoder-Decoder structure.
- **Self-Attention:** Understanding the core engine.
- **Responsible AI: Mitigating Bias & Hallucination.**

Phase 3: GenAI Engineering

- **Prompt Engineering:** Mastering Zero-shot, Few-shot, and Chain-of-Thought techniques for better outputs.
- **Fine-Tuning Strategies:** Learning Instruction Tuning and parameter-efficient methods like LoRA (PEFT).
- **Commercial APIs:** Hands-on experience with OpenAI (GPT), Google (Gemini), and Anthropic models.
- **Evaluation:** Quantitative and qualitative benchmarking of LLM outputs.



Phase 4: Advanced Systems

- **RAG Architecture:** Building Retrieval Augmented Generation using Vector DBs like Chroma and FAISS.
- **Agentic AI:** Implementing autonomous agents using the ReAct framework and LangGraph.
- **Tool Integration:** Connecting LLMs to external calculators, search engines, and custom APIs.
- **Deployment:** Bringing models to production with Streamlit and cloud infrastructure.

Questions?

Next Session: Lab 01 - Setting up the GenAI Environment